

SAAHITHI MUKALA

Threat Detection & Hunting | Detection Engineering | SIEM & Endpoint Telemetry

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SUMMARY

Final-year B.Tech Cybersecurity student (2027), Year-3 CGPA 8.0, ISC2 CC and CCNA certified, focused on threat hunting and detection engineering. Run a Wazuh SIEM homelab — measured detection coverage across 13 ATT&CK techniques and tuned paired-validated custom rules to cut false positives — and work hands-on with Microsoft Sentinel and KQL. Also built the evaluation harness for a team SOC-automation research project (paper in progress). Strong on SIEM alert triage, custom-rule tuning, endpoint / network / cloud / identity telemetry, MITRE ATT&CK and Python, with clearly documented findings and CTF work.

EDUCATION

B.Tech in Cybersecurity — Symbiosis Skills & Professional University, Pune

- **Year 3 CGPA: 8.0** • SGPA: 9.40 (Sem 6), 9.12 (Sem 5)

TECHNICAL SKILLS

- **Threat Detection & Hunting:** MITRE ATT&CK mapping, detection logic development & testing, alert triage, endpoint & network telemetry analysis, EDR, IOC analysis
- **SIEM / Security Platforms:** Microsoft Sentinel (deployment, data connectors/ingestion, analytics & detection rules, incident investigation), KQL / Kusto (security-log querying, operators & functions, log analysis for threat detection); Wazuh (SIEM deployment, log correlation, alert triage), Atomic Red Team adversary simulation, EDR tooling, detection-rule evaluation; TryHackMe SOC L1
- **Query & Scripting:** KQL (Kusto), SQL, Python (detection tooling, evaluation harnesses, automation), REST APIs; Git & GitHub, GitHub Actions CI/CD, detection-as-code
- **Endpoint & OS:** eBPF / kernel-level telemetry, Linux (Kali) administration, Wireshark packet analysis
- **Cloud & Infrastructure:** AWS (attack-path modelling, IAM, CSPM), Terraform (shift-left), Docker; NIST CSF, ISO 27001
- **Offensive & Assessment:** Metasploit, network security, cryptography, reverse engineering (CTF)
- **Development (secondary):** FastAPI, React, TypeScript, MongoDB

SECURITY PROJECTS

SOC Homelab — Detection Engineering & Custom Rule Development

Personal lab (ongoing)

Wazuh 4.14, Sysmon, Atomic Red Team, KVM/libvirt, Windows 11, PCRE2, MITRE ATT&CK — github.com/saahithimukala10-gif/soc-homelabs

- Deployed a Wazuh SIEM from scratch on an isolated KVM lab, instrumenting a Windows 11 endpoint with layered telemetry (Sysmon, Windows audit policy, PowerShell script-block logging) and validating the pipeline end-to-end from endpoint → agent → manager → indexer → dashboard.
- Ran **13 MITRE ATT&CK techniques across 6 tactics** via Atomic Red Team, measuring real out-of-box detection coverage with a telemetry-first methodology — confirming whether telemetry reached the SIEM before judging whether a rule fired.
- Authored and **paired-validated custom Wazuh rules** (PCRE2 regex, if_group / field / mitre tags) — including LSASS credential-dumping (T1003.001) and shadow-copy deletion / ransomware precursor (T1490) — each proven to fire on the attack and stay silent on a benign lookalike.
- Found and corrected real detection failures: a **stock LSASS rule firing on benign telemetry** (unanchored mask regex matching Defender's routine access while missing the actual dump) and a broad rule mis-tagging four unrelated techniques with a single hardcoded ATT&CK ID — reducing false positives and fixing misattribution.
- Diagnosed telemetry gaps upstream of the SIEM (disabled audit subcategories, Sysmon filters shipping ProcessAccess off, LSA Protection blocking capture) — detection engineering, not just rule writing.

End-to-End Open-Source SOC Pipeline for Automated Threat Detection, Intelligence Enrichment & Incident Response *Team research project*

Python, Ollama, Qwen 2.5 7B, Gemini, evaluation harness

- Owned the **evaluation harness** for a 4-person research project on prompt-injection resistance in automated SOC alert triage — logging every model verdict, tagging it correct / manipulated / false-negative against ground truth, and computing detection metrics as results streamed in.
- Ran **multi-model LLM inference** (Qwen 2.5 7B via Ollama on RTX 5060; Gemini via API), feeding each SOC alert both raw and through a defensive sanitization layer and recording triage verdicts per configuration.
- Tested against **poisoned MITRE ATT&CK alert variants** (hidden instructions in attacker-controlled fields) vs. clean baselines; quantified attack-success rate, triage precision/recall, false-negative rate and added latency across models.

- Consolidated results into the comparison tables underpinning the team's IEEE/Scopus-target research paper.

Kernox — eBPF Endpoint Detection & Response (EDR) Platform

Group project (PBL)

Python, eBPF, FastAPI, React, TypeScript

- Team-built an EDR platform monitoring Linux endpoints in real time via **eBPF kernel-level telemetry** with minimal overhead.
- Contributed the **detection-telemetry visualization layer** — a React/TypeScript dashboard surfacing alerts, endpoint insights and suspicious-behaviour signals from the backend correlation engine — and supporting backend API work.

CloudChain — Attack-Path-Aware Cloud Security Posture Management (AWS)

Team project (contributor)

Python, FastAPI, networkx, boto3, React — github.com/threethedrummer/CloudChain

- Modelled AWS resources as a **dependency graph to find realistic attack chains** (internet-reachable entry → AdministratorAccess) rather than isolated misconfigurations — an attacker-path view directly relevant to hunt hypotheses.
- Implemented read-only path validation (CONFIRMED / REFUTED / UNVERIFIABLE), context-aware risk scoring, Terraform shift-left and drift detection; backed by **194 tests** and GitHub Actions CI/CD.

CTF & LABS

Documented wargame writeups — github.com/saahithimukala10-gif/ctf-writeups

- **Bandit** (OverTheWire) — Linux/Unix fundamentals; **Natas** — web server-side vulns, auth flaws, injection; **Krypton** — cryptanalysis; **Leviathan** — reverse engineering & privilege escalation. **TryHackMe** (Top 9%) — SOC Level 1 & Pre-Security learning paths, plus hands-on challenge-room writeups (goal, recon, exploitation, privilege escalation).

EXPERIENCE

Software Development Intern — She Skillz Global

- Built cross-platform mobile apps (Flutter); ran functional, usability and reliability testing to improve stability.

Web Development & Testing Intern — Hamsa Hi-Tech

- Built responsive front-end features; executed functional/usability testing to find and resolve defects pre-release.

Data & Market Research Intern — CodeByte

- European market research and structured-database maintenance; contact sourcing via HubSpot / SalesQL.

CERTIFICATIONS & LANGUAGES

- **ISC2 Certified in Cybersecurity (CC)** • **Cisco CCNA** • TryHackMe (Top 9%): **Pre-Security, SOC Level 1** (In progress), Microsoft Sentinel, KQL (Kusto),
- Languages: English, Hindi, German (Basic)